



Traffic Signal Control Using Reinforcement Learning

Final Project Presentation

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The Problem with Fixed Timers

- Traffic jams waste time and fuel globally.
- The Problem: Fixed timers cannot adapt.
- Example: A light stays green for an empty street while cars pile up on a red light across the intersection.



Our Solution:

Reinforcement Learning (RL)

- What if the traffic light could learn from its mistakes?
 - Objective: Train an AI to minimize vehicle waiting time at a 4-way intersection.
 - No hard-coded rules: The AI learns entirely from scratch based on feedback.
-

Setting Up the Environment (MDP)



State

Number of cars waiting
North/South, East/West, and
current light color



Actions

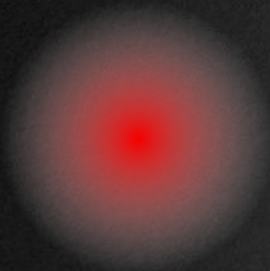
Keep light Green (0) or
Switch Light (1)



Dynamic Traffic:

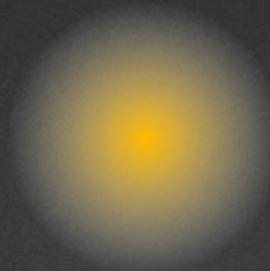
Simulates a Morning Rush,
Mid-day traffic, and an
Evening Rush.

The Three AI Algorithms We Tested



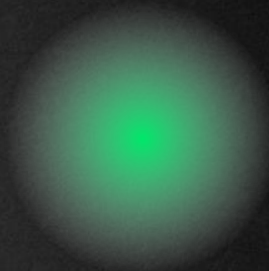
Q-learning

Off-policy. Very aggressive; assumes perfect future execution.



SARSA

On-policy. More cautious; takes its own random exploration mistakes into account.



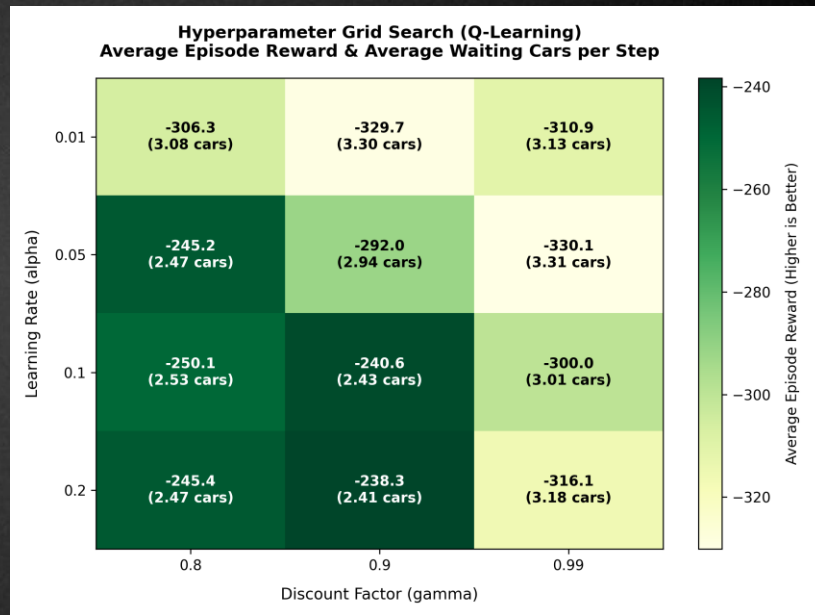
DQN

Uses a neural Network (6-> 64-> 64-> 2)
To handle complex patterns efficiently

Finding the best parameters

We tested extreme and moderate settings for the AI's learning

- Low Learning Rate = AI runs out of time to learn.
- High Discount Factor = AI looks too far into the future and gets confused.
- Result: Moderate settings (Alpha=0.1 , Gamma = 0.95) gave the best and most stable performance.



What is the Best Way to Grade the AI?

We tested 4 different scoring rules:

1. $R_t = -(q_{NS} + q_{EW})$

2. $R_t^{\text{max-queue}} = -\max(q_{NS}, q_{EW})$

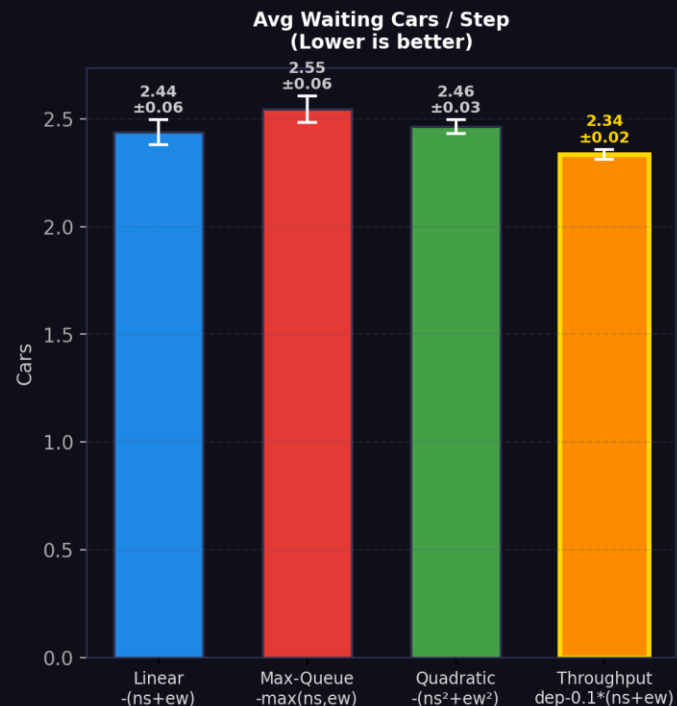
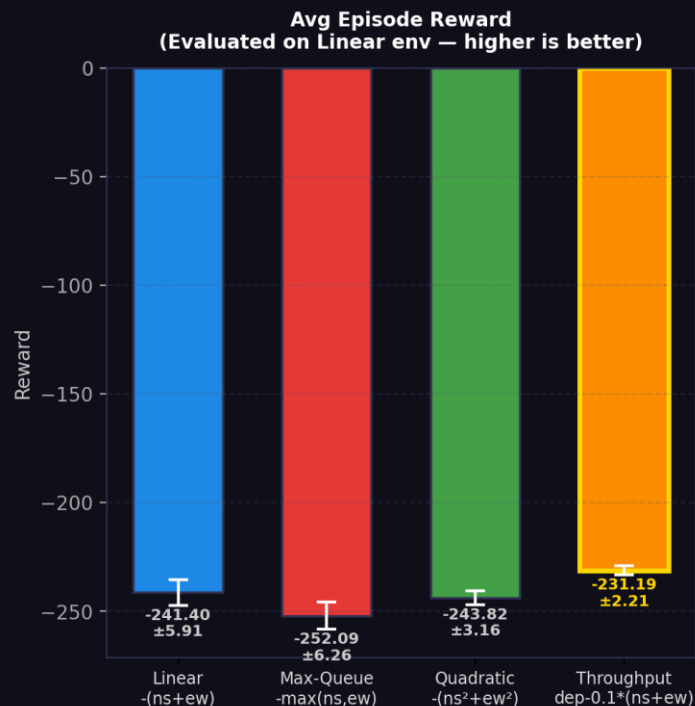
3. $R_t^{\text{quadratic}} = -(q_{NS}^2 + q_{EW}^2)$

4. $R_t^{\text{throughput}} = (d_{NS} + d_{EW}) - 0.1(q_{NS} + q_{EW})$

- d_{NS}, d_{EW} are the number of departures in the current step
- q_{NS}, q_{EW} are the number of cars waiting

- The Loser: "Max-Queue". It ignores half the intersection and confuses the AI.
- The Winner: "Throughput."

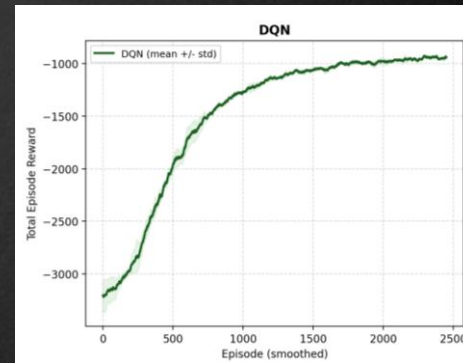
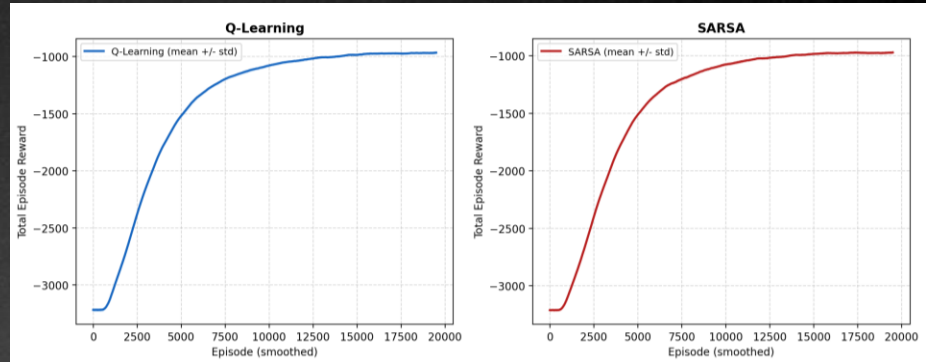
Reward Function Ablation Study — Final Evaluation (mean $\pm$ std, 5 seeds)



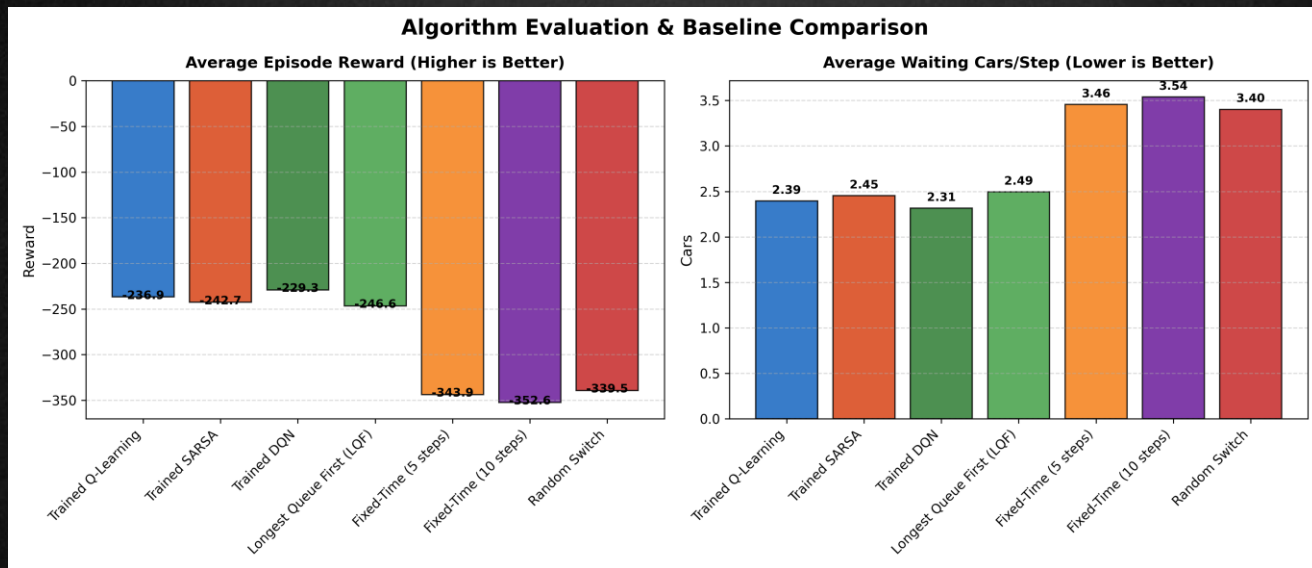
Training Results:

All three AI algorithms successfully learned!
And they improved their scores by 70% from their initial random guessing.

- DQN reached the absolute top score 12x faster than the tabular methods.



The Final Exam (Evaluating the AI)

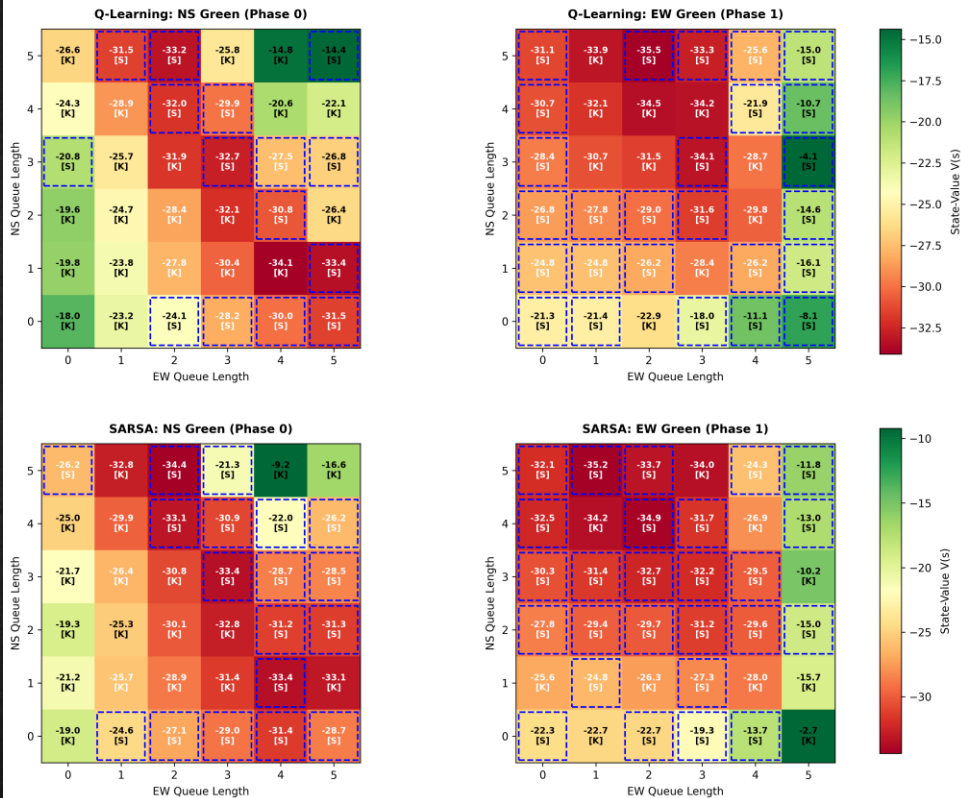


DQN was the overall winner: reduced wait times by 17% compared to the smart LQF rule, and improved 71% over fixed timers.

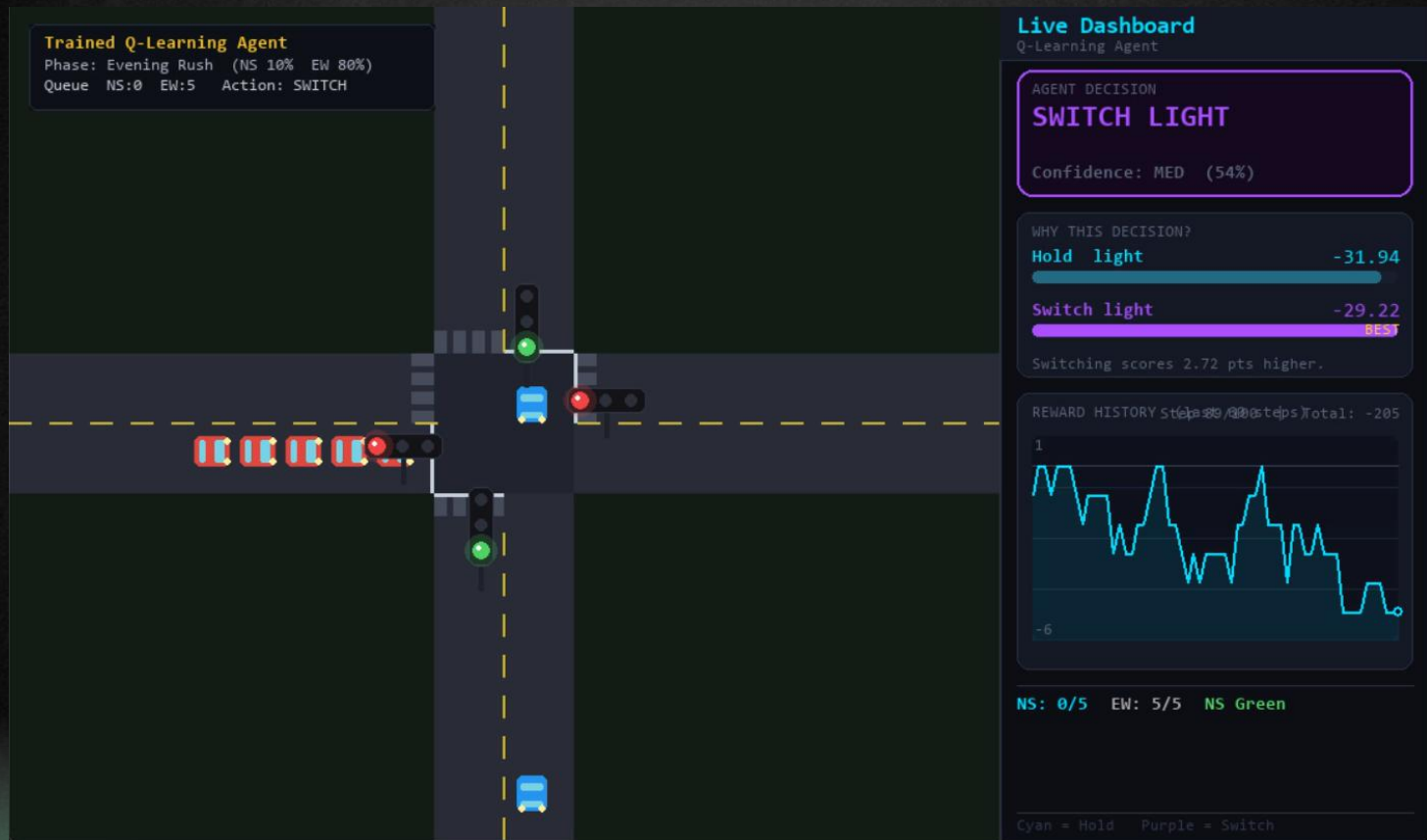
Looking Inside the AI's "Brain" :

Traffic Signal RL: Learned State-Value Functions V(s) & Decision Boundaries

Optimal Action: Switch Phase (a=1) [K] = Keep Green | [S] = Switch to Yellow/Other



Q-learning visualization



SARSA visualization:



Limitations & Future Work

Multi-agent RL
(Connecting a whole grid
of traffic lights to talk to
each other).



Our model uses a
simplified state (capped
at 15 cars) for an isolated
intersection.

Future Work :2
to explore a more
complex environment
with multiple
intersections.

Conclusions

01

AI consistently beats traditional, rule-based traffic controllers.

03

Focusing on moving cars (throughput) is a better strategy than just punishing traffic jams.



02

Neural Networks (DQN) are incredibly efficient at finding patterns quickly.

04

RL is a highly viable path toward solving modern city traffic!



THANK
YOU FOR
LISTINING!
